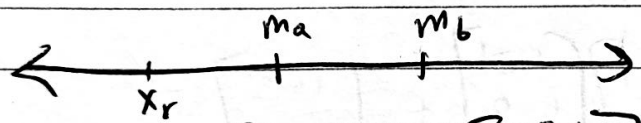
# Homework 7

S1-Q1) when a new landmark is found, you categorize it with robot pose & measurement to the mark. Assuming, it is outside of some mahabolis dist. add the state to $\mu$ + any correllation uncertainty for in relation to everything else

S1-Q2) landmark position is initialized using robot pose + measurement

S2-Q3) The $\Sigma_{mx}$, $\Sigma_{xm}$ & $\Sigma_{mm}$ are all updated

D1-Q1)




1-D robot

$x = [x_r, m_a, m_b]^T$

$\mu = \begin{bmatrix} 3 \\ 7 \\ 12 \end{bmatrix}$ $\Sigma = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 3 \end{bmatrix}$

measurements $z_1 = 12$, $z_2 = 11$

$\sim(0, 16)$ $\sim(0, 8)$

(assume no new landmark(s))

$H_j = m_j - x_r + \text{noise}$ $H_a = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$ $H_b = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$

$\hat{z}_a = 4$ $\hat{z}_b = 9$

$H_a \Sigma H_a^T = \begin{bmatrix} -1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 3 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$

$= \begin{bmatrix} -1 & 2 & -1 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$

$= 3$

$H_b \Sigma H_b^T = \begin{bmatrix} -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 3 \end{bmatrix} \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$

$= \begin{bmatrix} 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$

$= 2$

Nearest neighbor ($D^2 = v^2/s$)

$z_1 = 12$ $(\delta = 16)$

$v = 12 - 9 = 3$ $\quad s = 3 + 16 = 19$ $\quad D^2 = \frac{v^2}{s} = 9/19 = 0.5$

$v = 12 - 4 = 8$ $\quad s = 3 + 16 = 19$ $\quad D^2 = 64/19 = 3.37$

$z_2 = 11$ $(\delta = 8)$

$v = 11 - 9 = 2$ $\quad s = 2 + 8 = 10$ $\quad 4/10 = 0.4$

$v = 11 - 4 = 7$ $\quad s = 3 + 8 = 11$ $\quad 49/11 = 4.45$

bad, both go to $m_2$

b) joint compatability

$H = \begin{bmatrix} j_1 \\ j_2 \end{bmatrix}$ $\quad z_{jt} = h_{jt}(x_t^B) + w_{jt}$

$S_{jt} = H \Sigma H^T + \begin{bmatrix} 16 & 0 \\ 0 & 8 \end{bmatrix}$

$$\begin{bmatrix} -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 3 \end{bmatrix} \begin{bmatrix} -1 & -1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 16 & 0 \\ 0 & 8 \end{bmatrix}$$

$$= \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix} + \begin{bmatrix} 16 & 0 \\ 0 & 8 \end{bmatrix}$$

$$= \begin{bmatrix} 19 & 0 \\ 0 & 10 \end{bmatrix} \quad \text{or} \quad \begin{bmatrix} 19 & 0 \\ 0 & 11 \end{bmatrix}$$

$z_1 = 9$, $z_2 = 1$ $\quad v = [8, 2]$

$z_1 = 6$, $z_2 = 0$ $\quad v = [3, 7]$

$\det(S_1) = 190$ $\quad \det(S_2) = 198$

joint $D^2 = \frac{64}{19} + \frac{4}{10} = 3.77$ $\quad \frac{49}{18} + \frac{9}{18} = 4.95$

smaller so $z_1 = 9$, $z_2 = 6$